MILITARY SPECIFICATION

DETONATOR, STAB, M50
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-D-60093A(MU), dated 29 October 1973, and is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

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2. Drawings. Add the following:
"9298134 - Primer Mix (NOL #130)"

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4.4.1 Delete this paragraph in its entirety and substitute the following:

"4.4.1 Determination of moisture content of lead azide (see 4.3.3.1). A tared dried aluminum disn 2 inches in diameter shall be covered with a sheet of teflon 2 1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams shall be transferred to the moisture dish while both the cover and dish are on the balance pan. The dish and contents shall be transferred to an oven and dried at 95° plus or minus (±) 2°C for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The teflon cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{A-B}{A} \times 100$$

Where:

A = the weight of the specimen before drying.
B = the weight of the specimen after drying."

FSC 1375

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4.4.2 Primer mixture. Delete this paragraph in its entirety and substitute the following:

"4.4.2.1 Moisture content (see 4.3.3.2.1). Transfer an accurately weighed portion of approximately one gram of the primer mixture to a tared aluminum weighing dish provided with a teflon cover while both the dish and cover are on the balance pan. Place the dish and contents, with the cover removed, in an oven maintained at 60° ± 5°C for 2 hours. The cover shall be placed in a desiccator during the drying period. After drying, transfer the dish and contents quickly to a desiccator and cool at room temperature. Replace the plastic cover and weigh. Calculate the loss in weight of the dish cover and contents to percent moisture in the sample.

4.4.2.2 Composition (see 4.3.3.2.2). The analysis of the composition shall be performed as follows:

4.4.2.2.1 Barium Nitrate. Transfer an accurately weighed portion of approximately 0.5 grams of the dried sample (dried as described in 4.4.2.1) to a tared medium porosity fritted glass crucible of 30 milliliter (ml) capacity. Record the weight of crucible and sample as Weight A. Add 3 ml of distilled water which has been saturated with lead azide at ambient temperature. Swirl for 30 seconds, then apply suction. Repeat this procedure two times, making a total of three 3 ml extractions. Using an eye dropper, transfer to the crucible enough alcohol to thoroughly wet the sample and wash the crucible walls. Aspirate until the odor of alcohol is no longer detectable and dry the crucible in an oven for 30 minutes at 60° ± 5°C. Cool in a desiccator and weigh. Record this weight as Weight B. Reserve the crucible and residue for the normal or basic lead styphnate determination. Correct for the percentage of normal or basic lead styphnate in the water extract as follows: Transfer the filtrate and washings quantitatively to a 100 ml calibrated volumetric flask. Wash the beaker thoroughly with distilled water, add the washings to the flask, and dilute to the mark of the volumetric. Using a calibrated pipette, transfer a 10 ml aliquot of this solution to a second 100 ml calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell (see 6.3) having a width of approximately 1 centimeter (cm) and determine its optical density at a wave length of 410 millimicrons. Use as reference a similar cell containing the same proportion of water saturated with lead azide and alcohol as the sample solution.

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Since the cell which holds the lead styphnate solution is not identical with the reference cell, correct the optical density obtained for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with a solution of water and alcohol, and measure at a wave length of 410 millimicrons, the optical density of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal or basic lead styphnate in the filtrate on a dry basis as follows:

Normal lead styphnate found in barium nitrate filtrate,
Percent $P_n = \frac{2.94\ (A-B)}{WD}$

Basic lead styphnate found in barium nitrate filtrate,
Percent $P_b = \frac{4.48\ (A-B)}{WD}$

where:

A = Optical density of normal lead styphnate solution.
B = Optical density of water and alcohol solution.
D = Width of corex cell, cm.
W = Weight of dried sample.

Calculate the percentage of barium nitrate in the sample as follows:

Barium nitrate, percent $= \frac{100\ (A-B)}{W} - (P_b \text{ or } P_n)$

where:

W = Weight of the dried sample.
A = Weight of crucible and sample.
B = Weight of the crucible and sample after extractions with water saturated with lead azide.
P_n = Percentage of normal lead styphnate found in barium nitrate filtrate.
P_b = Percentage of basic lead styphnate found in barium nitrate filtrate.

4.4.2.2.2. Basic or normal lead styphnate. The residue obtained in the determination of barium nitrate shall be extracted with six 5ml portions of saturated ammonium acetate solution at about 25°C, (hot ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each

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portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water, saturated with antimony sulfide, until the washing coming through the crucible appears colorless. Reserve the crucible and residue for the tetracene determination. The filtrate and washings shall be quantitatively transferred to a 250 ml calibrated volumetric flask. Wash the beaker thoroughly with distilled water, adding the washings to the flask and dilute to the mark.

Using a calibrated pipette, a 2 ml aliquot of this solution shall be transferred to a 50 ml calibrated volumetric flask in the case of basic lead styphnate, or a 2 ml aliquot of the solution shall be transferred to a 100 ml calibrated volumetric flask in the case of normal lead styphnate, and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of approximately one centimeter (cm) and its optical density determined at a wave length of 410 millimicrons. As a reference, a similar cell containing ammonium acetate solution of the same concentration as the solution of the specimen shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the optical density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. This shall be done by filling both cells with ammonium acetate solution and measuring at a wave length of 410 millimicrons, the optical density of the cell which originally contained the lead styphnate solution. The percentage of normal or basic lead styphnate in the specimen shall be calculated as follows:

$$\text{Percentage basic lead styphnate} = \frac{28\ (A-B)}{WD} + P_b$$

$$\text{Percentage normal lead styphnate} = \frac{36.75\ (A-B)}{WD} + P_n$$

where:

A = Optical density of normal or basic lead styphnate
B = Optical density of ammonium acetate solution.
D = Width of corex cell, cm.
W = Weight of dried specimen.
P_b = Percent basic lead styphnate found in barium nitrate filtrate.
P_n = Percent normal lead styphnate found in barium nitrate filtrate.

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4.4.2.2.3 Tetracene. The wet residue obtained in the determination of basic or normal lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 ± 5°C, for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml beaker with a stream of water, 25 ml of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 ± 5°C for 30 minutes, cooled in a desiccator and weighed. The percentage of tetracene in the specimen shall be calculated as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W}$$

where:

C = Weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = Weight of crucible and residue after treating with hot water.
W = Weight of the dried specimen.

4.4.2.2.4 Antimony sulfide. The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the specimen as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)\ 100}{W}$$

where:

D = Weight of crucible and specimen after treating with hot water.
E = Weight of crucible.
W = Weight of dried specimen.

4.4.2.2.5 Lead azide. The lead azide content in the specimen shall be determined by subtracting from 100 the combined percentage of barium nitrate, basic or normal lead styphnate, tetracene and antimony sulfide."

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6.4 Change "SARPA-QA-A-M" to "DRDAR-QAR-R" and "SARPA-AD-E-P-Z" to "DRDAR-LCE-D".

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-A206)

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